

IINTS-AF Physiology And Data Brief

Competition evidence bundle for EUCYS review

Project: IINTS-AF SDK

Purpose: Explain why realistic glucose simulation is difficult and why data quality matters.

Audience: EUCYS judges, biology/medicine reviewers, interdisciplinary science reviewers.

This PDF was rendered from the markdown report and packaged benchmark artifacts inside the IINTS-AF SDK workspace.

Executive Summary

Type 1 diabetes is difficult to simulate because glucose is affected by food, insulin, exercise, sensor lag, sleep, stress, illness, and individual physiology. A curve that is smooth or mathematically convenient can still be unrealistic. IINTS-AF therefore treats physiology and data quality as first-class parts of the SDK.

The SDK does not claim to model a human perfectly. It claims to make the assumptions visible, testable, and comparable.

Type 1 Diabetes In One Paragraph

In type 1 diabetes, the body cannot produce enough insulin to regulate blood glucose. Insulin lowers glucose by helping move glucose from the blood into tissues and by influencing glucose production. Meals raise glucose as carbohydrates are absorbed. Exercise can lower glucose and can change insulin sensitivity. Because insulin action is delayed and food absorption is variable, safe dosing is a prediction problem under uncertainty.

This is why insulin algorithms need more than a clean control loop. They need realistic disturbances, delays, edge cases, and safety checks.

Why Real Glucose Curves Are Messy

A realistic CGM trace is not just a sine wave or a smooth meal spike. It can contain:

Physiological or data feature	Why it matters
Delayed insulin action	Insulin given now affects glucose later
Meal absorption variability	The same carbohydrate amount can produce different curves
Exercise effects	Activity can increase glucose use and hypoglycemia risk
Sensor lag	Interstitial CGM glucose can lag blood glucose
Noisy measurements	CGM devices can show artifacts or dropouts
Individual differences	Insulin sensitivity and carbohydrate ratio vary across people

Physiological or data feature	Why it matters
Overnight drift	Glucose can rise or fall without a visible meal event
Stacked insulin	Multiple boluses can overlap and increase low-glucose risk

A simulator that ignores these features may pass a demo but fail the scientific question.

CGM Metrics Used In The SDK

IINTS-AF uses standard CGM interpretation metrics because they are understandable and widely used in diabetes research.

Metric	Meaning
Time in Range	Percent of time between 70 and 180 mg/dL
Time Below Range	Percent of time below 70 mg/dL
Time Below 54	Percent of time in clinically more severe hypoglycemia
Time Above Range	Percent of time above 180 mg/dL
Time Above 250	Percent of time in marked hyperglycemia
Mean glucose	Average glucose level
Coefficient of variation	Relative glucose variability

The international Time in Range consensus and ADA Standards support this style of CGM interpretation. IINTS-AF uses these metrics for research evaluation, not as individual treatment advice.

Why Time In Range Is Not Enough

A high Time in Range score can hide different risk profiles. For example, two algorithms may both reach good TIR, but one may produce more hypoglycemia or require many more supervisor interventions.

That is why IINTS-AF reports multiple dimensions together:

- TIR for overall performance
- time below 70 and below 54 for low-glucose risk
- time above 180 and above 250 for hyperglycemia burden
- intervention counts for safety-supervisor burden
- worst-case events for stress-test interpretation

This prevents a single number from becoming misleading.

Why Data Quality Is A Scientific Issue

Machine learning and simulation pipelines often fail quietly when the data is bad. In diabetes, bad data can include impossible glucose jumps, missing insulin events, meal annotations that do not match the curve, duplicated timestamps, sensor dropouts, and corrupted units.

That is why MDMP exists inside the SDK. It checks whether a dataset looks usable before it becomes evidence.

The final EUCYS benchmark supports this point: clean certified conditions outperformed corrupted uncertified conditions by 17.11 Time-in-Range points. That does not prove every dataset problem, but it shows why trustworthy inputs matter.

Why Simulation Is Still Useful

Simulation is not a replacement for clinical evidence. It is a pre-clinical filter. It helps researchers ask:

- Does an algorithm behave sensibly before real-patient testing?
- What happens under rare or dangerous edge cases?
- Does the safety layer catch risky outputs?
- Does performance remain stable across patient profiles and random seeds?
- Which scenarios expose weaknesses?

This follows a broader biomedical simulation principle: models are simplified representations used to study systems that are difficult, expensive, or unsafe to test directly at first.

Why IINTS-AF Looks At Real Data Literature

The SDK source legend includes public and peer-reviewed references because synthetic traces need anchors. Examples include:

Source family	Why it matters for IINTS-AF
ADA Standards of Care 2026	Current clinical framing for diabetes goals and technology
International Time in Range consensus	Standard CGM metric interpretation
OhioT1DM dataset	Real CGM, insulin, meal, exercise, and life-event data from people with T1D
Dalla Man / Cobelli meal models	Physiological basis for meal-glucose-insulin simulation
UVA/Padova simulator literature	Established in-silico T1D simulator lineage
2024 generative T1D simulator paper	Modern evidence that simulator realism remains an active research problem

The Nature Communications Medicine Paper

The 2024 Communications Medicine paper on generative deep learning for T1D simulation is important because it supports the exact concern behind this SDK: classical physiological simulators can miss parts of real glucose-insulin complexity. The paper trained generative models using T1D datasets including OhioT1DM and evaluated whether generated patients showed realistic glycemic metrics and causal relationships between insulin, carbohydrates, and glucose.

For IINTS-AF, the takeaway is not "replace physiology with AI." The takeaway is stronger:

Realistic diabetes simulation is hard enough that both physiology and data-driven validation matter.

What The SDK Should Improve Next

The next scientific improvements should be:

- more real-data realism dashboards comparing synthetic and real CGM distributions
- better meal-response validation against public datasets
- documented sensor artifact scenarios
- explicit unit tests for physiological plausibility ranges
- transparent labels separating simulated, synthetic, public, and private data

Physiological Claim

The strongest physiological claim is:

IINTS-AF does not pretend that a virtual patient is a real patient. It makes virtual-patient assumptions inspectable and tests algorithms across physiology-inspired disturbances before any real-world claim is made.

Sources

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